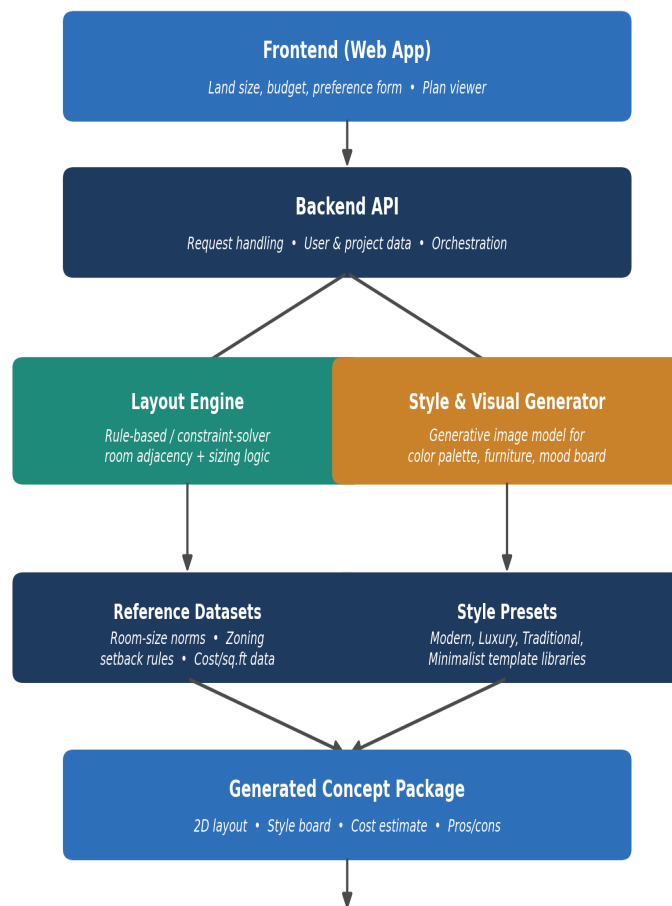


AI House Planner & Interior Designer

AI-Powered Home Design Concept Generator

Project Documentation — Revised Edition

System Architecture



⚠ Conceptual output only — recommend licensed architect review before construction

Introduction

I want to build an AI-powered website that helps people design their dream home based on the size of their land, budget, and personal preferences.

Problem Statement

When people build a house, they usually depend on engineers and architects. However, many homeowners struggle to visualize the best layout for their available land. Poor space planning can make the house feel congested, and mismatched interiors, wall colors, lighting, and furniture can reduce the overall beauty and uniqueness of the home.

Goal

The website should provide users with a clear visual concept of their future home, helping them make better design decisions before consulting an architect. It should combine smart space planning with aesthetically matching interiors to create a beautiful, functional, and personalized home design.

Proposed Solution (Revised)

An intelligent web application where users enter their land dimensions, budget, number of floors, family size, and design preferences. The system generates a conceptual home design — positioned clearly as a starting point for planning, not a construction-ready blueprint.

What the system generates

- Optimized 2D room layout based on land size (rule-based, not freeform AI guessing)
- Balcony, kitchen, bedroom, bathroom, and parking placement
- Space utilization suggestions to minimize congestion
- Interior design recommendations: wall colors, lighting, furniture, décor
- Rough cost estimate based on selected region and material grade
- Pros and cons of the generated design
- Multiple design styles: Modern, Luxury, Traditional, Minimalist, and more

Important correction: true architectural floor-plan generation (structurally sound, code-compliant, to-scale) cannot be fully automated by AI alone. This system generates a conceptual layout using a rule-based space-planning engine, paired with AI-generated interior style visuals — and every output carries a disclaimer recommending review by a licensed architect before construction.

User Workflow

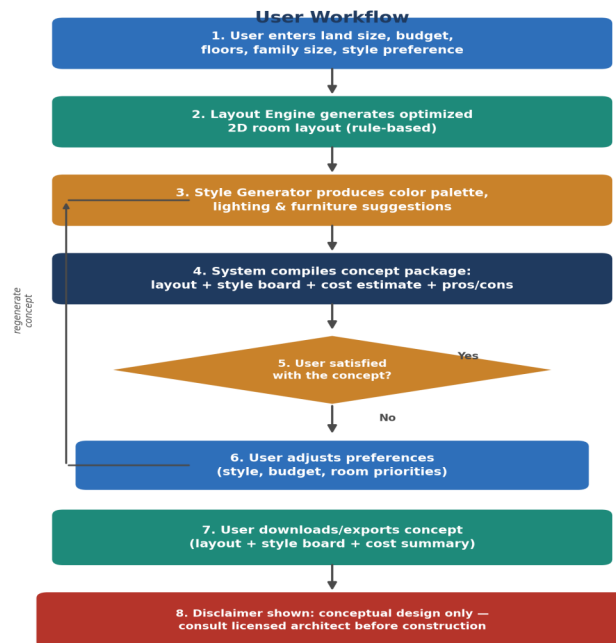


Figure 1 — End-to-end user workflow, including regeneration loop

Functional Requirements

1. User Input

- User registration & login
- Enter land dimensions (sq ft), budget, number of floors
- Enter family size and room priorities
- Select design style preference (Modern, Luxury, Traditional, Minimalist, etc.)
- Select region (for cost and code-norm context)

2. Layout Generation

- Rule-based room layout engine (adjacency, sizing, natural light/ventilation rules)
- Room placement: bedroom, kitchen, bathroom, balcony, parking
- Space utilization scoring to flag congestion

3. Interior & Style Recommendations

- AI-generated wall color palette suggestions
- Ceiling lighting and furniture suggestions
- Home décor recommendations matched to selected style
- Style board / mood board visual output

4. Output & Review

- Pros and cons summary of the generated design
- Rough cost estimate (region-adjusted)
- Regenerate / adjust preferences loop
- Downloadable concept package (layout + style board + cost summary)
- Clear disclaimer: conceptual design, not a construction blueprint

Non-Functional Requirements

Attribute	Requirement
Performance	Layout generation should return a concept within a few seconds for typical land sizes.
Reliability	Generated layouts must always satisfy basic room-size and adjacency rules — no overlapping or unusably small rooms.
Scalability	System should support many concurrent design requests.
Usability	Simple, visual-first UI usable by non-technical homeowners.
Accuracy Disclaimer	All outputs clearly labeled as conceptual — not construction-ready blueprints.

Attribute	Requirement
Localization	Cost estimates and room norms should be adjustable per region/country.
Data Security	Protect user project data (land details, budget, personal preferences).
Availability	Core layout + style generation should work reliably; heavy 3D rendering can degrade gracefully on slow connections.

System Architecture

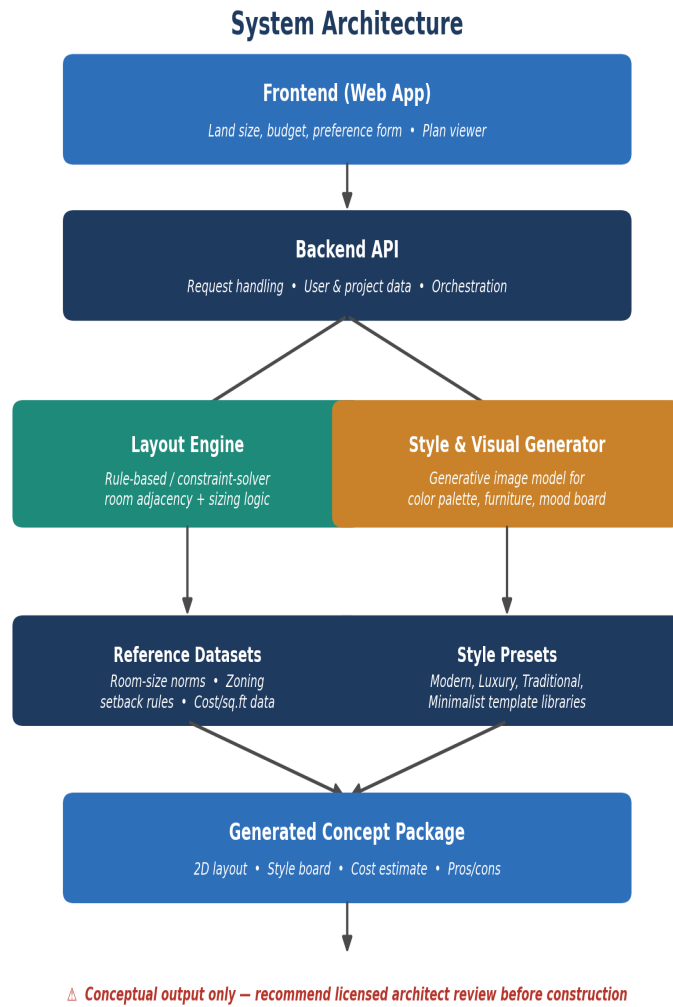


Figure 2 — Layered system architecture, separating layout logic from style generation

Technologies Used (Suggested)

- Frontend: React / Next.js with an interactive layout viewer (canvas or SVG-based)
- Backend: Node.js or Python (FastAPI) for orchestration
- Layout Engine: Constraint-solver or rule-based algorithm (e.g. graph-based room adjacency logic)
- Style/Visual Generator: Generative image model (e.g. Stable Diffusion) constrained by style presets
- Database: PostgreSQL or MongoDB for user/project data
- Reference Data: Room-size norms, regional cost-per-sq-ft tables, zoning notes (informational only)

Real-World Corrections & Improvements

The following changes were made to the original design to make it viable and responsible outside a controlled demo environment:

Issue in Original Plan	Real-World Fix
"Generate a complete house plan" implies a structurally sound, buildable blueprint — AI cannot guarantee this.	Reposition as a conceptual design visualizer. Add a clear disclaimer: outputs are a starting point, not a substitute for a licensed architect/structural engineer.
Pure text-to-image generation for floor plans gives inconsistent, non-to-scale layouts.	Use a rule-based / constraint-solver layout engine for room placement and sizing; use generative image models separately for interior style visuals only.
Layout generated from square footage alone ignores local building codes.	Scope to "general best-practice layout"; note that setback rules, FSI, and zoning vary by city/country and are out of scope for v1.
Budget-to-construction-cost assumed to be a single global formula.	Build a per-region cost database (cost per sq ft varies by location and material grade); let users select their region.
Full 3D interior rendering planned for v1.	Start with 2D top-down layout + AI-generated style boards (palette, furniture, lighting); move 3D rendering to v2.
No mention of dataset/reference source for room-size norms.	Use established space-planning norms and, if available, public floor-plan datasets (e.g. research datasets like RPLAN) to ground the rule engine.
No liability framing for a home-design output.	Legal disclaimer on every generated plan; recommend architect consultation before construction.

Project Roadmap

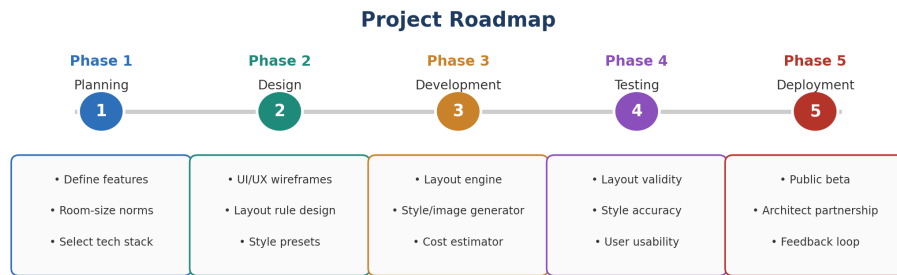


Figure 3 — Five-phase delivery roadmap

Suggested MVP scope: build only the 2D layout engine + style/color recommendations first (core of Phase 3). Push full 3D rendering and precise, code-compliant planning to a later phase once the core concept-generation loop is validated with real users.